

Crib Sentry

The baby monitor that alerts instead of only recording.

Tier 2

Seth Alvarez

ITAI 1378: Computer Vision Final

github.com/SalvarezJ/Crib-Sentry

The problem

A traditional baby monitor is only useful when someone is watching it. Parents cannot watch all the time and could miss much of what the camera records.

The goal is to cut down on what they miss.

What the system reports

All clear

Left the crib

Not visible

Hazard present

Who cares

Parents and caregivers of infants and toddlers.

Why it matters

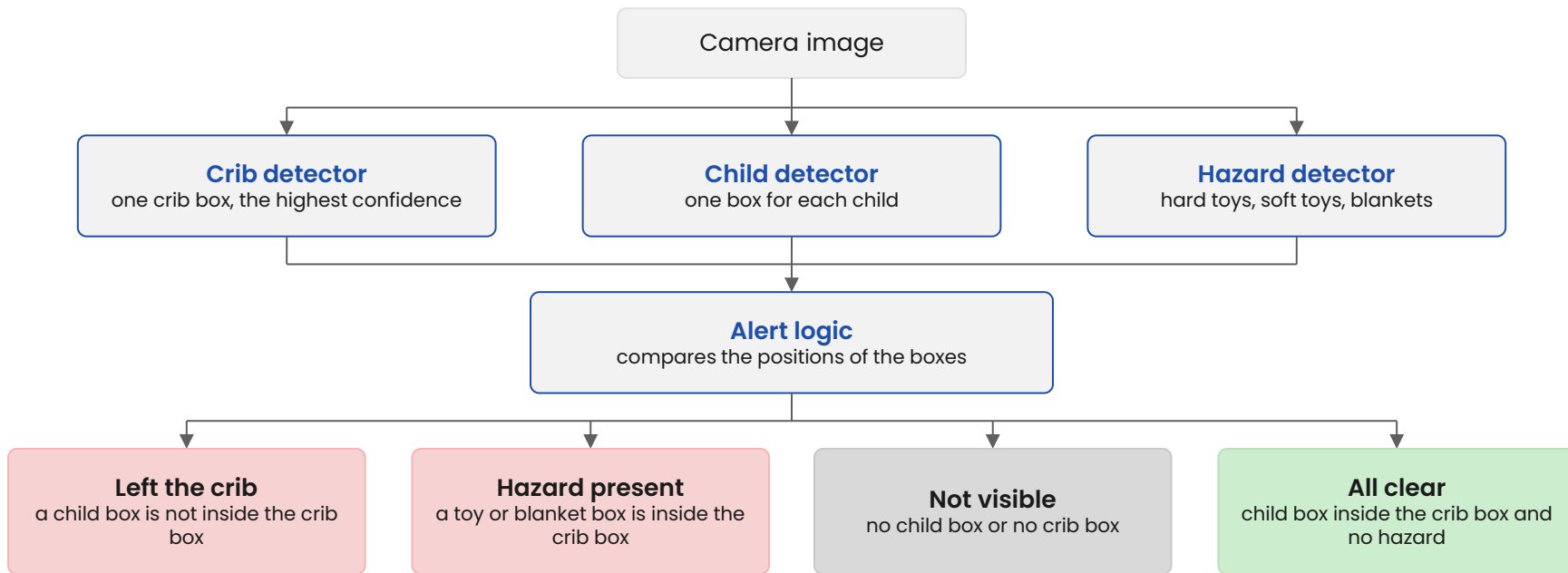
A camera that no person watches gives no protection.

The solution

A system that looks at the image and gives an alert when one of its conditions is true.

What I built

One image from a crib camera goes in, three detectors run, and my alert logic compares the boxes and gives one of four alerts.

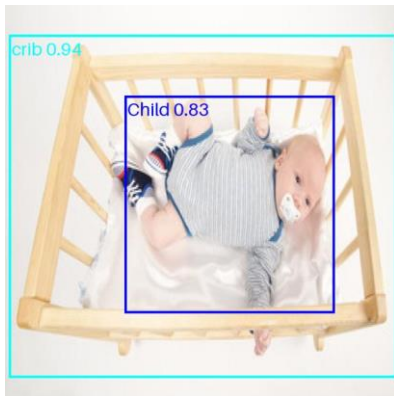


The order of the alerts is from left to right. With no crib box the alert is always not visible.

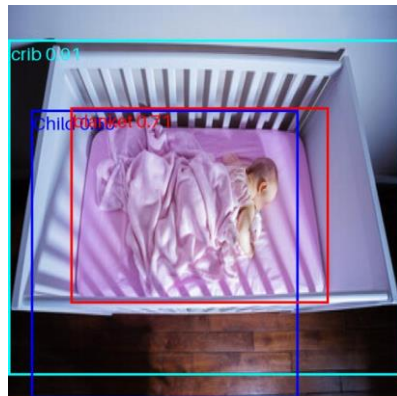
My Blueprint has two detectors. The system now has three.

Demo

Four correct alerts from my demo notebook. Cyan is the crib box, blue is the child box and red is a hazard box.



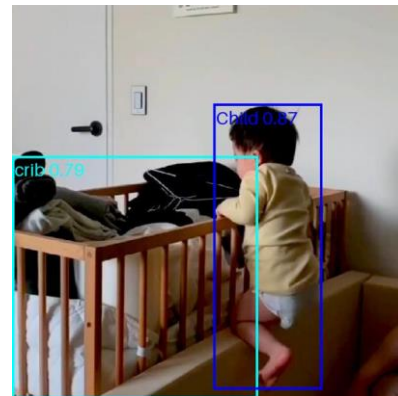
All clear
crib 0.94 · child 0.83



Hazard present
crib 0.91 · blanket 0.71



Not visible
crib 0.87 · no child box



Left the crib
crib 0.79 · child 0.87

Demo video: https://drive.google.com/file/d/1M_8wIFlzOgJ_6HV5io6qun-LyD5dzt_a/view

Notebook: notebooks/09_demo.ipynb in my repository

How it works

CV TECHNIQUE

Object detection

Each detector draws boxes and gives a confidence number from 0 to 1.

ARCHITECTURE AND MODEL

CNN: YOLO11n, three models

One finds the crib, one finds the child and one finds toys and blankets.

HOW I USE IT

Transfer learning

Each model starts from COCO weights and trains for 50 epochs on my images.

FRAMEWORK

PyTorch through Ultralytics

Free and open source.

Training runs on a free Colab T4 GPU.

The alert logic. `fraction_inside` divides the overlap of two boxes by the area of the inner box. It gives 1 when the box is fully inside and 0 when the boxes do not touch. A box is inside the crib when the result is more than 0.5.

During training I also add darker, lighter, blurred and rotated copies of the images so that the models do better on real rooms.

The data I used

Source	Used for	Size	Labels
Baby in Crib	Crib and Child training, and my 24 test images	242 images (169 / 49 / 24)	Child, Crib
Crib_detection	Crib training	1,494 images	crib, as outlines
BASE crib only BABY only	Child training	309 original images	baby
Baby object detection final	Child training, and my 10 "left the crib" test images	3,136 original frames	baby
CribHD T and B	Hazard training	1,369 training images	hard-toy, soft-toy, blanket
CribHD-C	Test only	120 images	None. I tag each image myself.

Cleaning and labels

I remove 18 wrong crib labels and I change outlines into boxes. I keep one file for each original image and I move 101 copies to the training folder.

Test images stay separate

No detector trains on a test image. I remove five videos from the Child training data because they show or can show my test rooms.

Roboflow datasets: CC BY 4.0. CribHD: non-commercial. Cost: \$0.

Results vs my targets

ALERT ACCURACY · TARGET NOT MET

70.8%

17 of 24 test images. My target is 85% or more.

TIME FOR EACH IMAGE · TARGET MET

0.034 s

on a T4 GPU. My target is less than 1 second.

Metric	My Blueprint target	What I achieved
Alert accuracy on the 24 test images	85% or more	70.8% (17 of 24)
Time for each image on a T4 GPU	Less than 1 second	0.034 seconds
"Left the crib" on 10 images from a different dataset	No target	7 of 10. The first system gets 0.
Alert accuracy on the 120 CribHD-C images	No target	41.7% (50 of 120). The first system gets 16.7%.

Why CribHD-C is low. The first system is my Blueprint design, with one detector for the child and the crib. It finds the crib in 22 of 120 images. The new Crib detector finds it in 76, and that takes the accuracy from 16.7% to 41.7%. The child is found in 104, so the missed crib is the main cause and not the blanket.

What changed from the Blueprint

My Blueprint	What I build	Why
Two detectors: one for the child and the crib, one for hazards	Three detectors: Crib, Child and Hazard	The first detector fails on real rooms: 0 of 10 on "left the crib"
Two datasets: Baby in Crib and CribHD	Three more datasets for crib and child training	Baby in Crib contains only studio photos
Labels used as they are	18 wrong crib labels removed	They put a crib box on a baby with no crib
Risk 2 has a Plan B: find a small extra dataset	10 "left the crib" test images, and five videos removed from training	No first dataset has a child out of a crib

What stays the same

Object detection, YOLO11, transfer learning, PyTorch through Ultralytics, the four alerts, the two metrics with their targets, and a cost of \$0.

Challenges and fixes

1. The first system gets 0 of 10 on "left the crib"

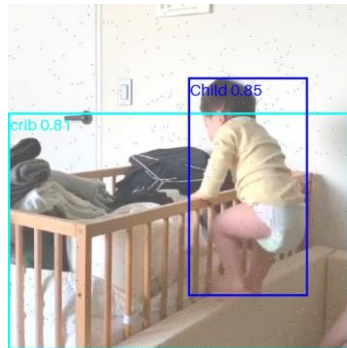
Solution: I train a Crib detector and a Child detector on real rooms, and the system gets 7 of 10. The cost is one alert on my 24 test images; 17 are correct now and 18 before.

2. A side view misleads my "inside" rule

The child climbs out, but more than half of the child box is still on the crib box. Not fixed: the rule works when the camera is above the crib.

3. The Crib detector is the weakest part

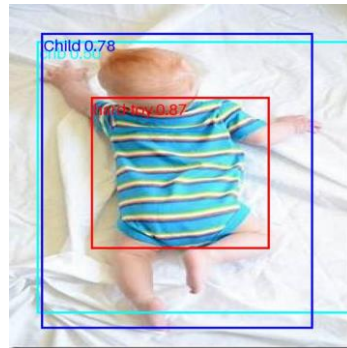
It puts a weak crib box on photos with no crib, and it finds a crib in only 76 of 120 CribHD-C images. Not fixed: this would be my next step.



Side view

Alert: all clear

Correct: left the crib



No crib in the photo

Alert: hazard present

Correct: not visible

Other errors on the 24 test images. The Hazard detector reads two fitted sheets as blankets. The Child detector does not find one baby.

Lessons and next steps

MY GREATEST LESSON

My Blueprint never expected the risk that the system does not find the crib.

104

of 120 CribHD-C images: child found

76

of 120 CribHD-C images: crib found

Three of my four alerts need a crib box, so each missed crib is a wrong alert.

What I build next

1. Judge the safety of the child with no crib box

With no crib box the system gives "not visible", also when it finds a hazard. That works against my own order of alerts.

2. A minimum confidence for the crib box

I select it on validation images so that a weak crib box does not start the other rules.

Repository: github.com/SalvarezJ/Crib-Sentry